

```
import java.io.*;
import java.util.*;
interface PerformOperation {
    boolean check(int a);
}
class MyMath {
    public static boolean checker(PerformOperation p, int num) {
        return p.check(num);
    }
}
```

// Write your code here

```
public static PerformOperation isOdd() {
    return i -> {
        return i % 2 != 0 ? true : false;
    };
}
```

```
public PerformOperation isPrime() {
    return i -> {
        if (i == 2) return true;
        if (i <= 1 || i % 2 == 0) return false;

        int numberSqrt = (int) Math.sqrt(i);

        for (int j=3; j<=numberSqrt; j+=2) {
            if (i % j == 0) return false;
        }

        return true;
    };
}
```

```
};  
}
```

```
public static PerformOperation isPalindrome() {  
    return i -> {  
        String numberStr = String.valueOf(i);  
        String reversedStr = "";  
  
        for (int j=numberStr.length()-1; j>=0; j--) {  
            reversedStr += numberStr.charAt(j);  
        }  
  
        return numberStr.equals(reversedStr);  
    };  
}  
}
```

```
public class Solution {  
  
    public static void main(String[] args) throws IOException {  
        MyMath ob = new MyMath();  
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));  
        int T = Integer.parseInt(br.readLine());  
        PerformOperation op;  
        boolean ret = false;  
        String ans = null;  
        while (T--> 0) {  
            String s = br.readLine().trim();
```

```
StringTokenizer st = new StringTokenizer(s);
int ch = Integer.parseInt(st.nextToken());
int num = Integer.parseInt(st.nextToken());
if (ch == 1) {
    op = ob.isOdd();
    ret = ob.checker(op, num);
    ans = (ret) ? "ODD" : "EVEN";
} else if (ch == 2) {
    op = ob.isPrime();
    ret = ob.checker(op, num);
    ans = (ret) ? "PRIME" : "COMPOSITE";
} else if (ch == 3) {
    op = ob.isPalindrome();
    ret = ob.checker(op, num);
    ans = (ret) ? "PALINDROME" : "NOT PALINDROME";

}
System.out.println(ans);
}
}
}
```